The Thirsty Cloud

Water supply and cooling choices for Singapore's data-centre growth

DSS5202 Sustainable Systems Analysis
2027–2035 planning scenarios

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1. Problem and comparison basis

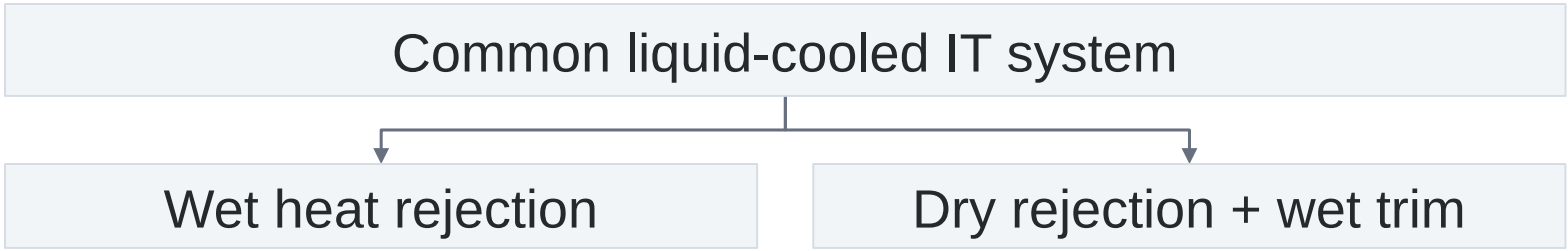
Additional water demand from new data centres during 2027–2035

Planning scenarios: 200, 500 and 700 MW by 2035

Functional unit: heat rejection for 1 MWh of IT energy

Partial boundary: water supply and incremental heat rejection

2. Four strategies and a common baseline



A1 Desalination
A2 NEWater
A3 NEWater + tower reuse

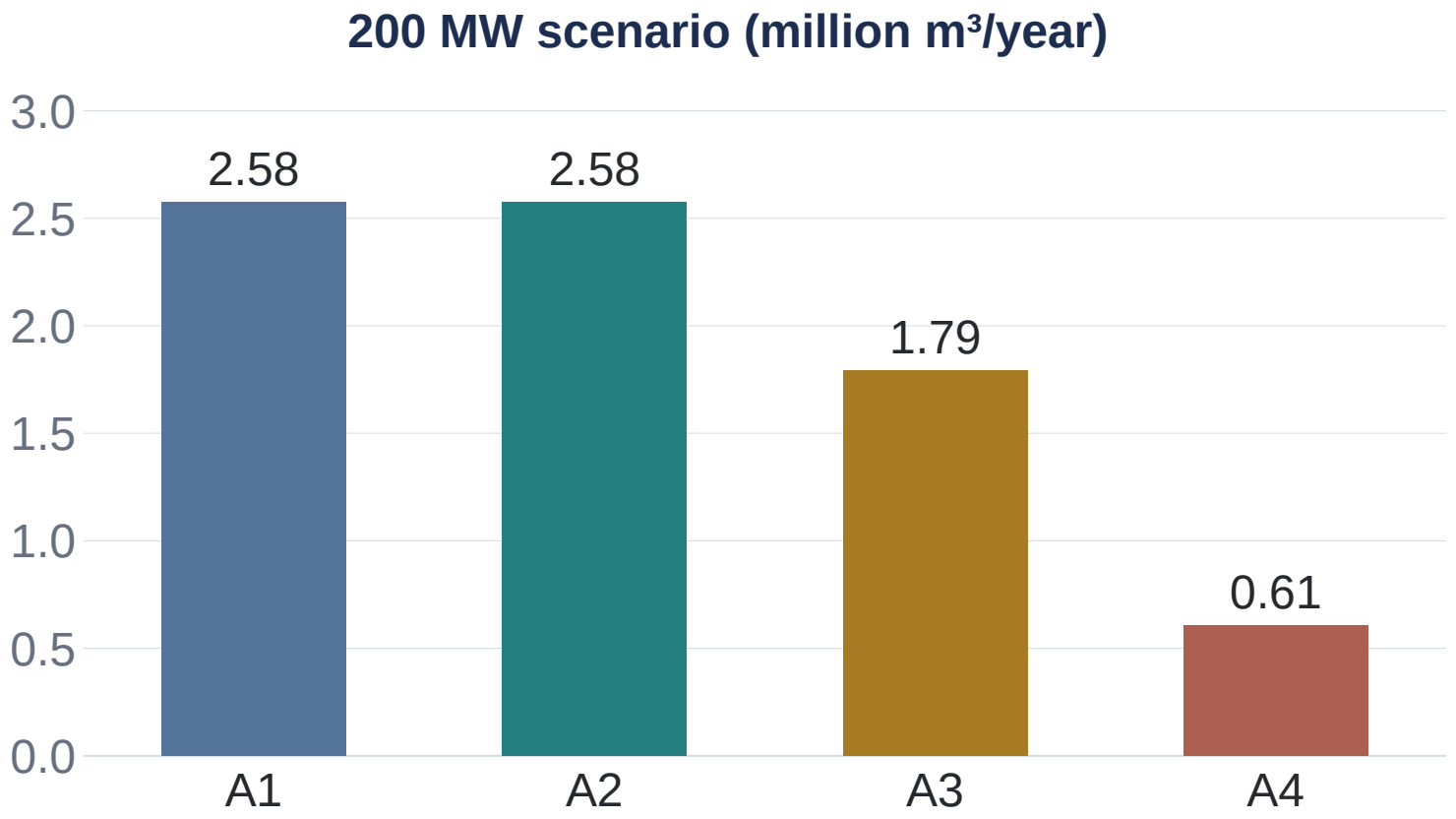
A4 NEWater for remaining demand

Liquid-cooled IT in all strategies

A4 has no separate liquid-cooling credit

Incremental dry-rejection electricity and capital cost are included

3. Water demand in 2035

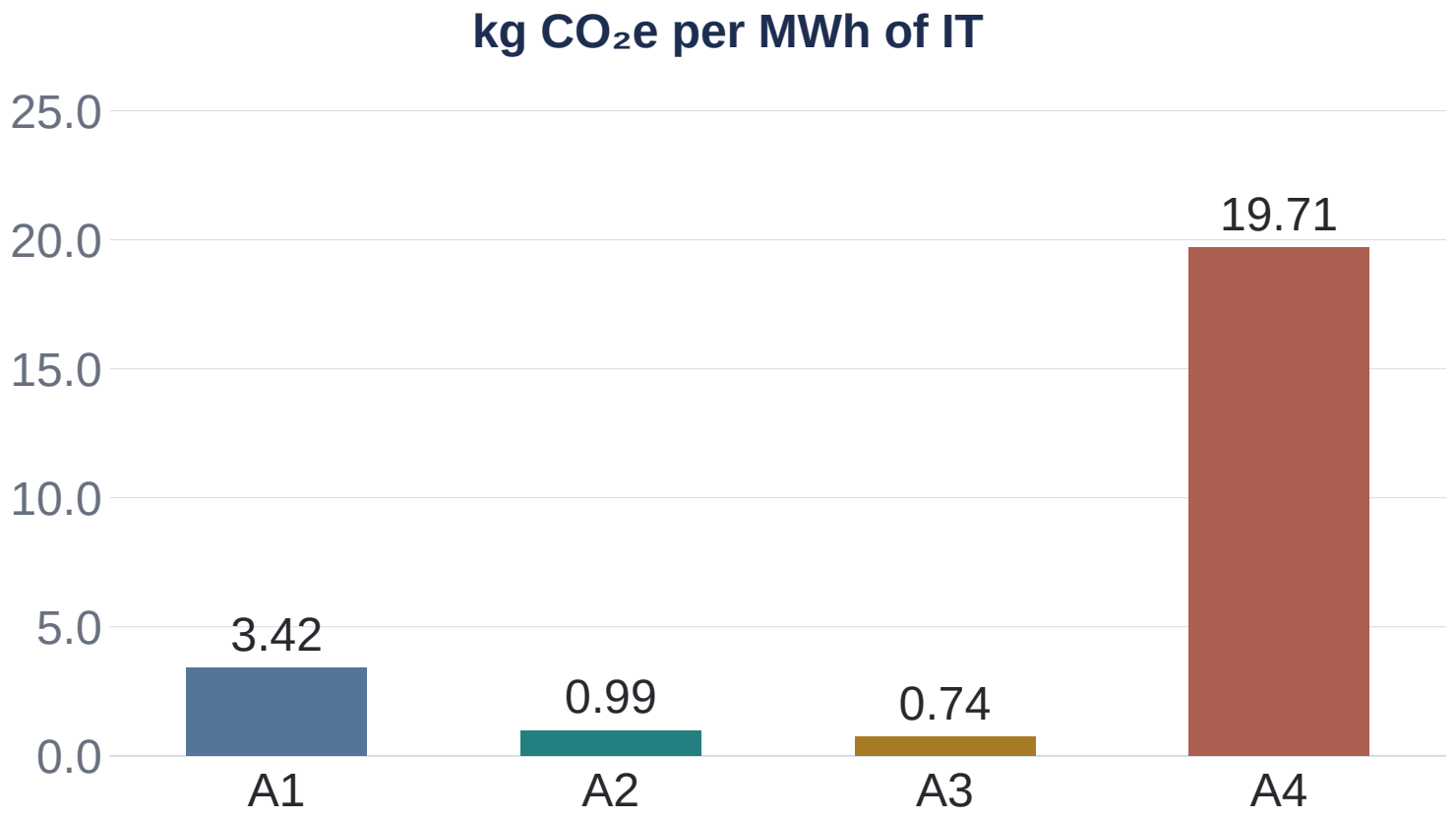


A3 water reduction
30% relative to A2

A4 water reduction
76% relative to A2

Historical weather is a screening input, not equipment validation

4. Greenhouse-gas comparison



Tower reuse has the lowest base emissions within the model boundary

Water savings and electricity savings need separate assessment

Common IT and liquid-loop emissions are excluded

5. Social and operator costs

Cost per MWh of IT	A1	A2	A3	A4
Social cost (S$)	2.60	1.00	1.54	11.60
Operator cost (S$)	6.80	5.25	4.83	15.64

NEWater has the lowest base social cost.

Tariffs belong in the operator calculation.
Carbon-price cases are reported separately from financial costs.

6. Broader priorities and AHP

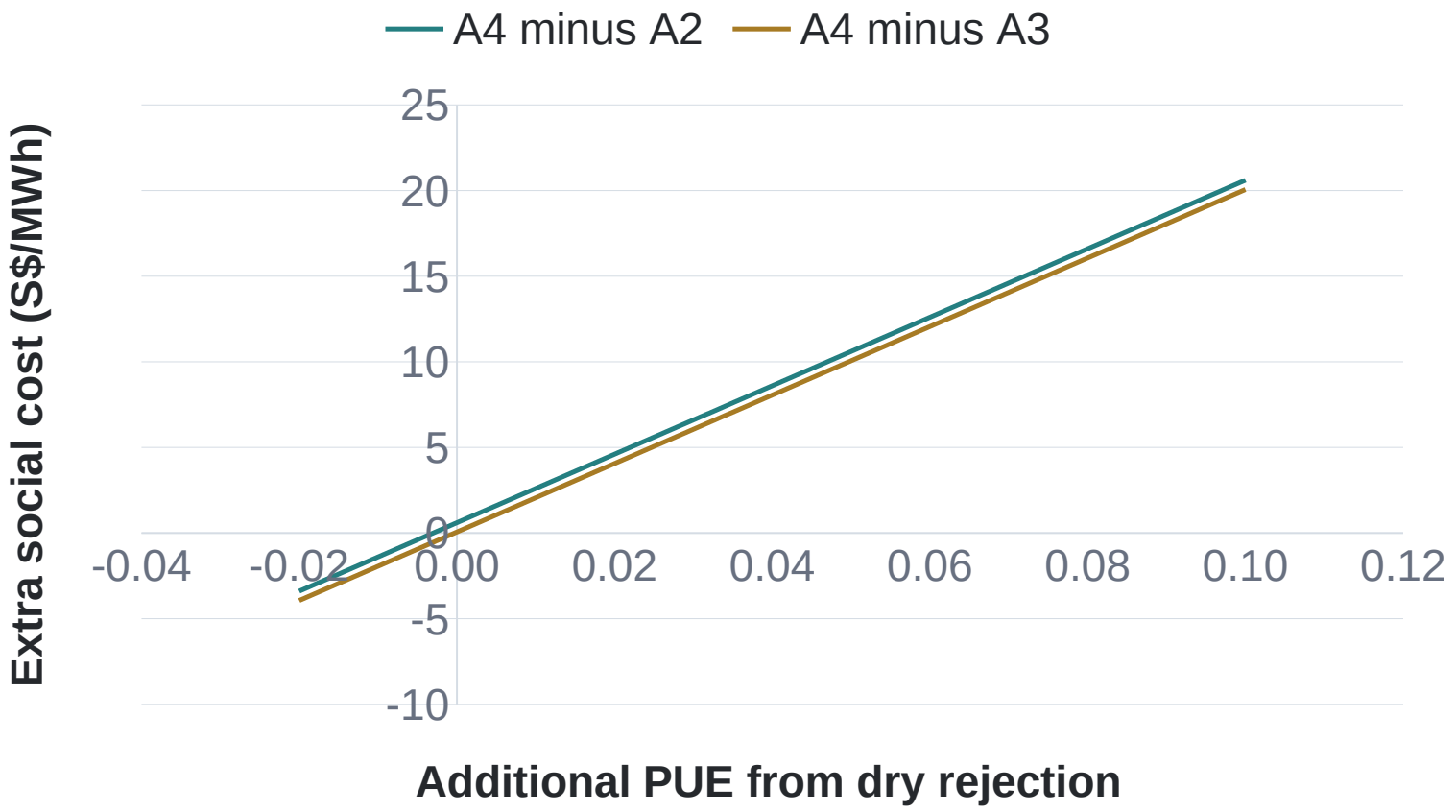
Balanced scenario	A1	A2	A3	A4
Priority	0.231	0.250	0.299	0.220
Rank	3	2	1	4

Weights represent illustrative priorities, not measured stakeholder preferences.

Sensitivity checks vary weights and reduce overlap between criteria.

Consistency ratios assess internal consistency, not evidence quality.

7. Electricity and cost sensitivity



A4 minus comparator cost

Above zero: A4 costs more

Base dry-rejection capital cost held constant

Preference uncertainty is tested separately

8. Recommendations and limitations

PUB: compare the energy required for each unit of water saved.

IMDA and operators: assess WUE and PUE together under actual loads.

Tower reuse needs water-quality and cost checks. Dry rejection needs site performance evidence.

Treat these results as a screening comparison with explicit assumptions.

9. Conclusion

Dry rejection uses the least water.
NEWater has the lowest social cost.

The preferred strategy depends on electricity use, equipment cost and decision priorities.

Model assumptions and supporting calculations accompany the report.